Falcon 9 Landing Prediction

From launch data to explainable recovery predictions

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Executive Summary

GOAL

Predict whether the first stage of a SpaceX Falcon 9 rocket will land successfully.

METHODS

Data collection, wrangling, SQL, EDA, Folium, Dash-style analytics and classification.

VERIFIED DATA

90 launch records, 18 variables and 83 encoded model features.

RESULTS

60 successful landings out of 90 (66.7%).

Best reproduced test accuracy: 83.3%.

Introduction & Business Problem

WHY LANDINGS MATTER

Reusable first stages can lower launch costs. Predicting recovery outcomes helps understand the mission conditions linked to successful landings.

RESEARCH QUESTION

How do launch site, payload mass, orbit and booster features relate to landing success?

TARGET

Class = 1: successful landing

Class = 0: unsuccessful/other outcome

Data Collection: SpaceX API & Web Scraping

COURSE WORKFLOW

- SpaceX REST API provides launches, payload and booster details.
- Wikipedia launch tables can be scraped using requests and BeautifulSoup.
- API and scraping notebooks were supplied as source templates.

DATA USED FOR THIS REPORT

- IBM Skills Network dataset_part_2.csv
- 90 Falcon 9 launch records (2010–2020).
- The API and scraping notebooks supplied were not executed; no live API or Wikipedia results are claimed here.

Data Wrangling & Feature Engineering

CLEANING

- Standardized outcome into binary Class.
- Inspected payload mass, orbit, launch site and mission date.
- Dataset used for analysis has 90 rows and 18 columns.

FEATURES

- dataset_part_3.csv: 90 rows × 83 predictors.
- Orbit, LaunchSite, LandingPad and Serial encoded as indicator columns.
- Numerical features include PayloadMass, Flights, Block and ReusedCount.
- Training-only standardization prevents test leakage.

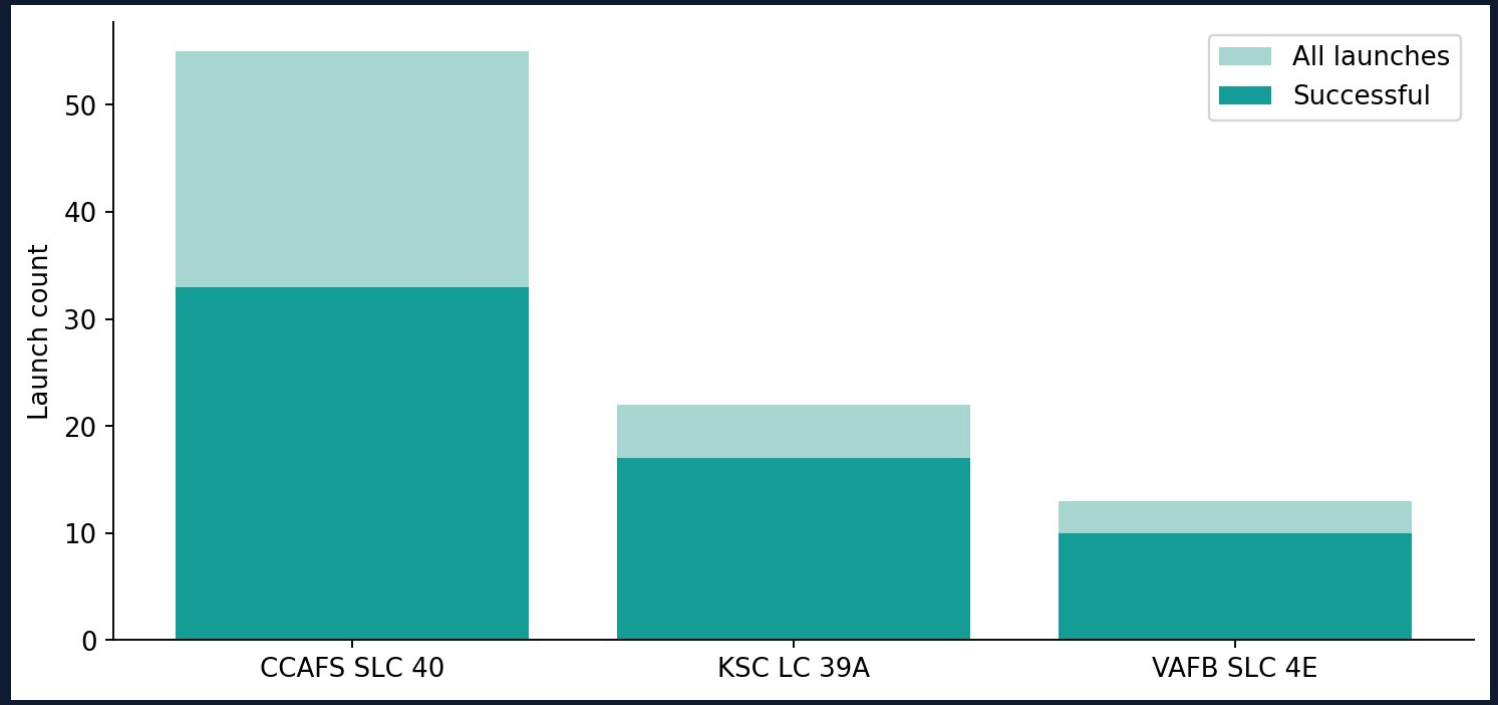
EDA: Launch Sites

Launches by site

CCAFS SLC 40: 55 launches, 33 successful

KSC LC 39A: 22 launches, 17 successful

VAFB SLC 4E: 13 launches, 10 successful



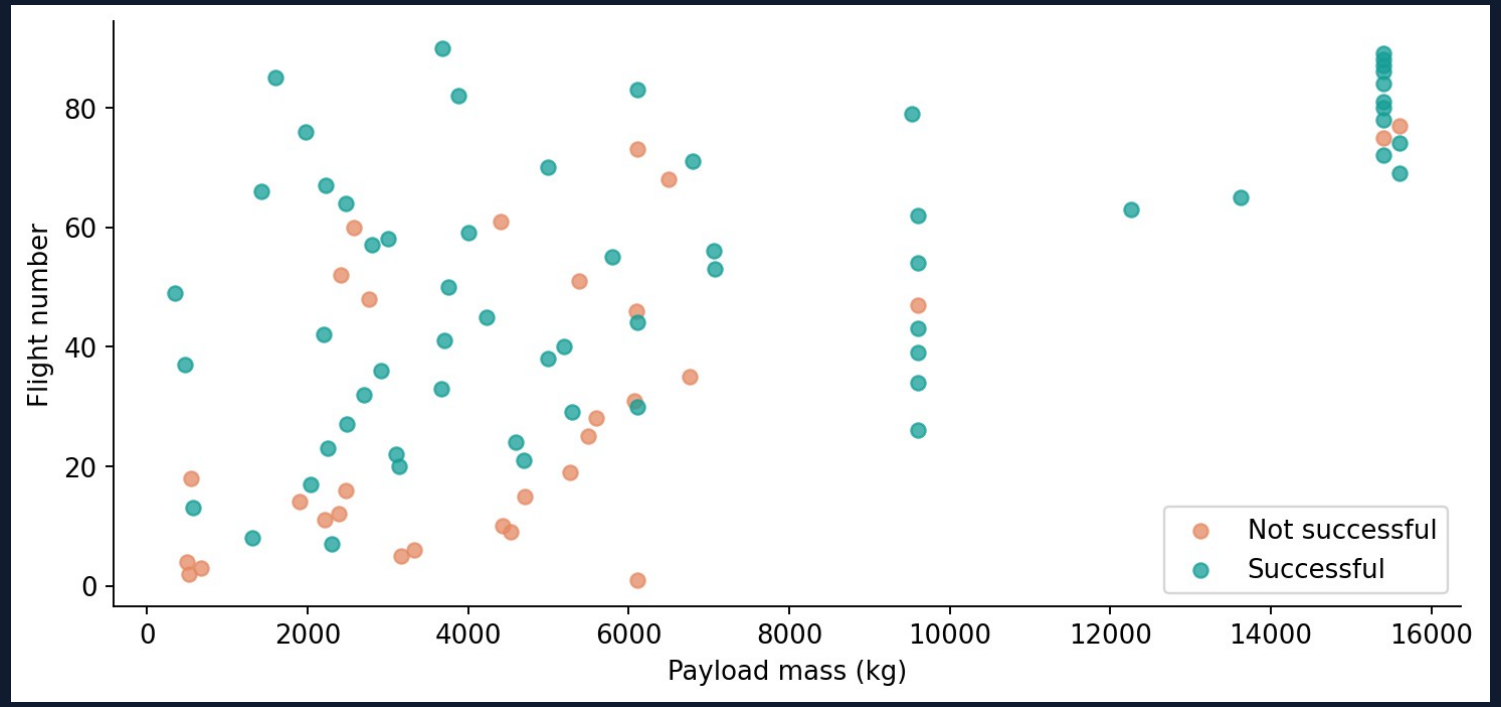
Computed directly from dataset_part_2.csv

EDA: Payload Mass & Flight Number

Each dot is one launch.

Green indicates a successful landing; orange indicates other outcomes.

This visual explores the relationship between payload mass and flight sequence.

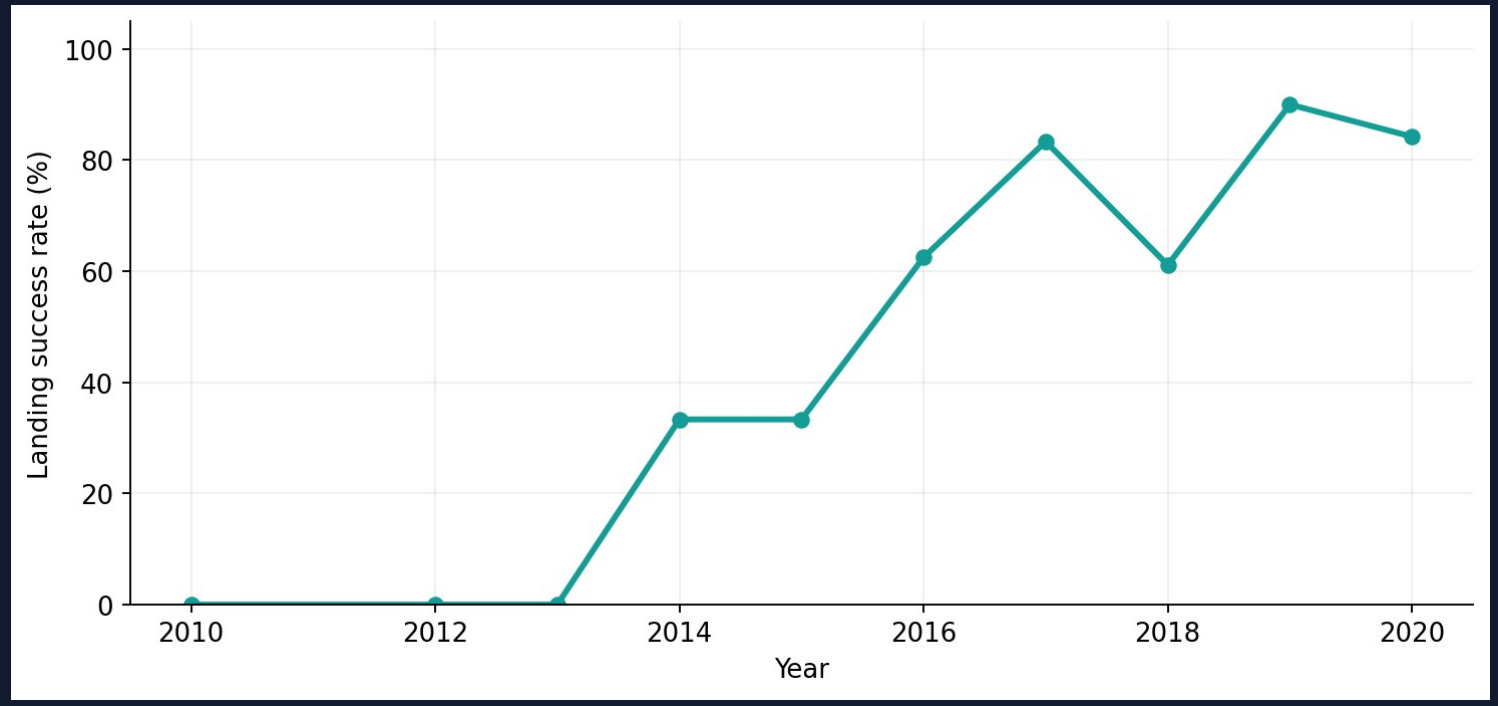


Source: dataset_part_2.csv

EDA: Landing Success Over Time

Annual success rate is the share of launches with Class = 1.

Historical improvement is visible, but small annual sample sizes can make rates volatile.



Calculated from the Date and Class fields

EDA & SQL: Orbit-Level Comparison

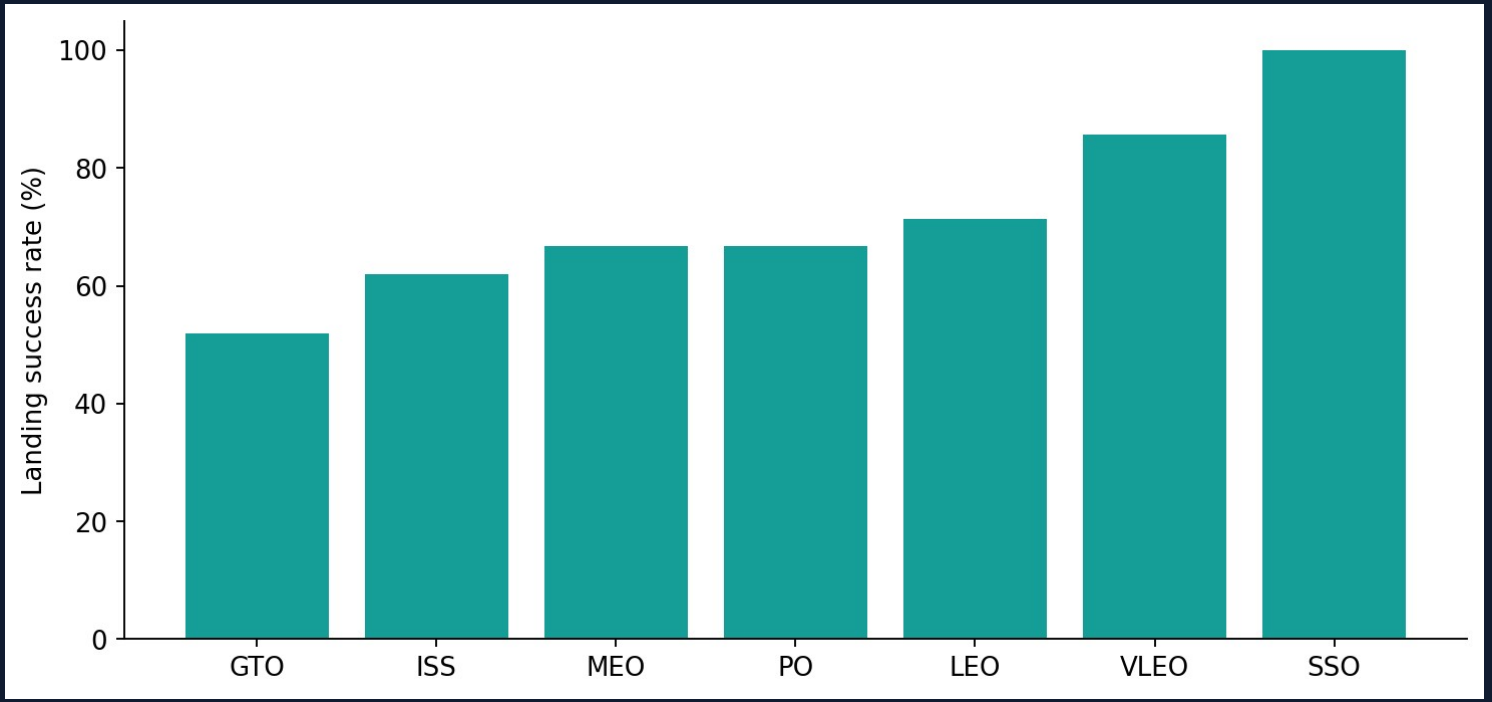
SQL-style analysis

GROUP BY Orbit

COUNT(*) AS launches

AVG(Class) AS success_rate

The chart shows orbits with at least three recorded launches.



Aggregated from the supplied IBM CSV

SQL Findings: Sites & Outcomes

SITE RANKING

CCAFS SLC 40: 55 launches; 60.0% successful

KSC LC 39A: 22 launches; 77.3% successful

VAFB SLC 4E: 13 launches; 76.9% successful

LANDING OUTCOMES

True ASDS: 41

None None: 19

True RTLS: 14

False ASDS: 6

True Ocean: 5

False Ocean: 2

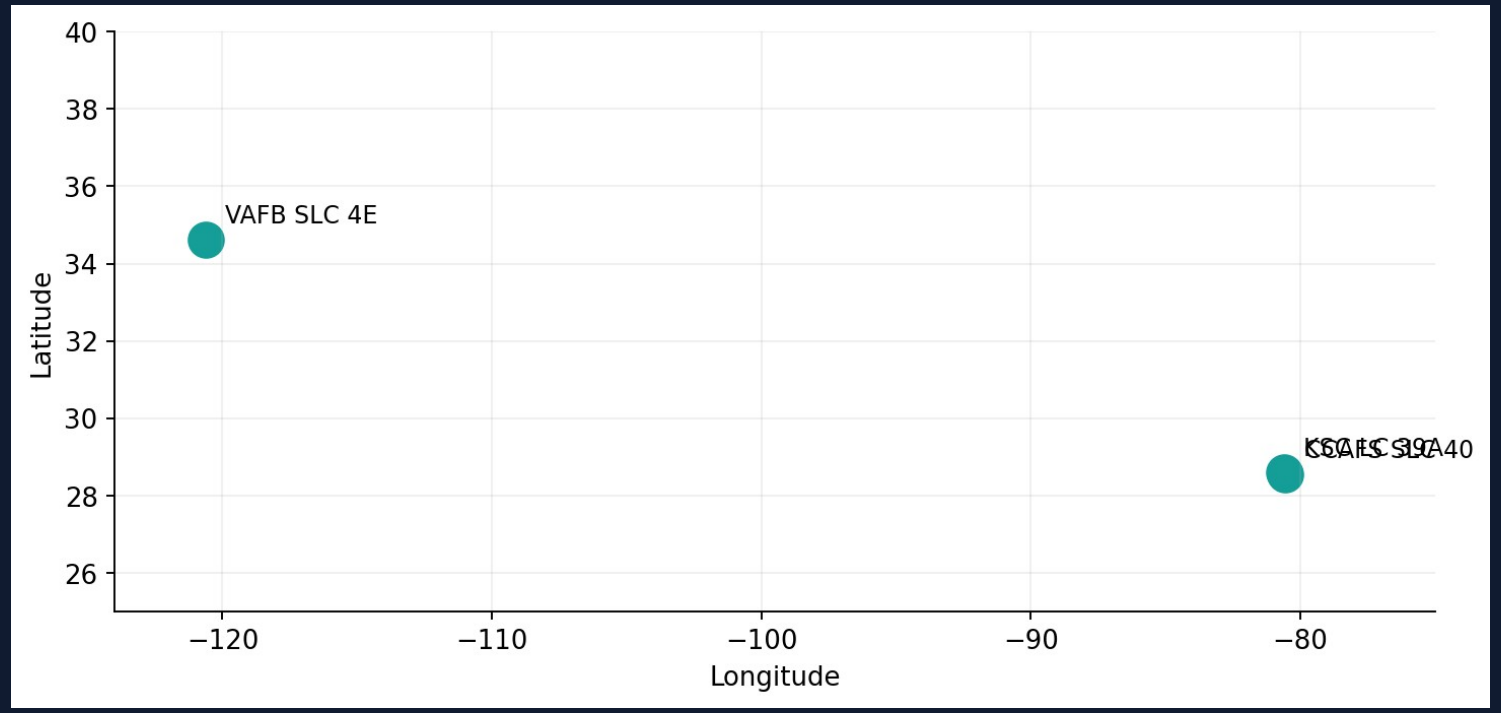
Results were reproduced using pandas aggregation, equivalent to SQL GROUP BY logic.

Folium: Launch-Site Locations

Location coordinates come from Latitude and Longitude fields.

The provided Folium lab describes site markers and proximity to coastlines, highways and railways.

Exact proximity distances were not computed from the provided CSV.



Coordinate plot derived from dataset_part_2.csv; not a Folium screenshot

Interactive Analytics: Dash Concepts

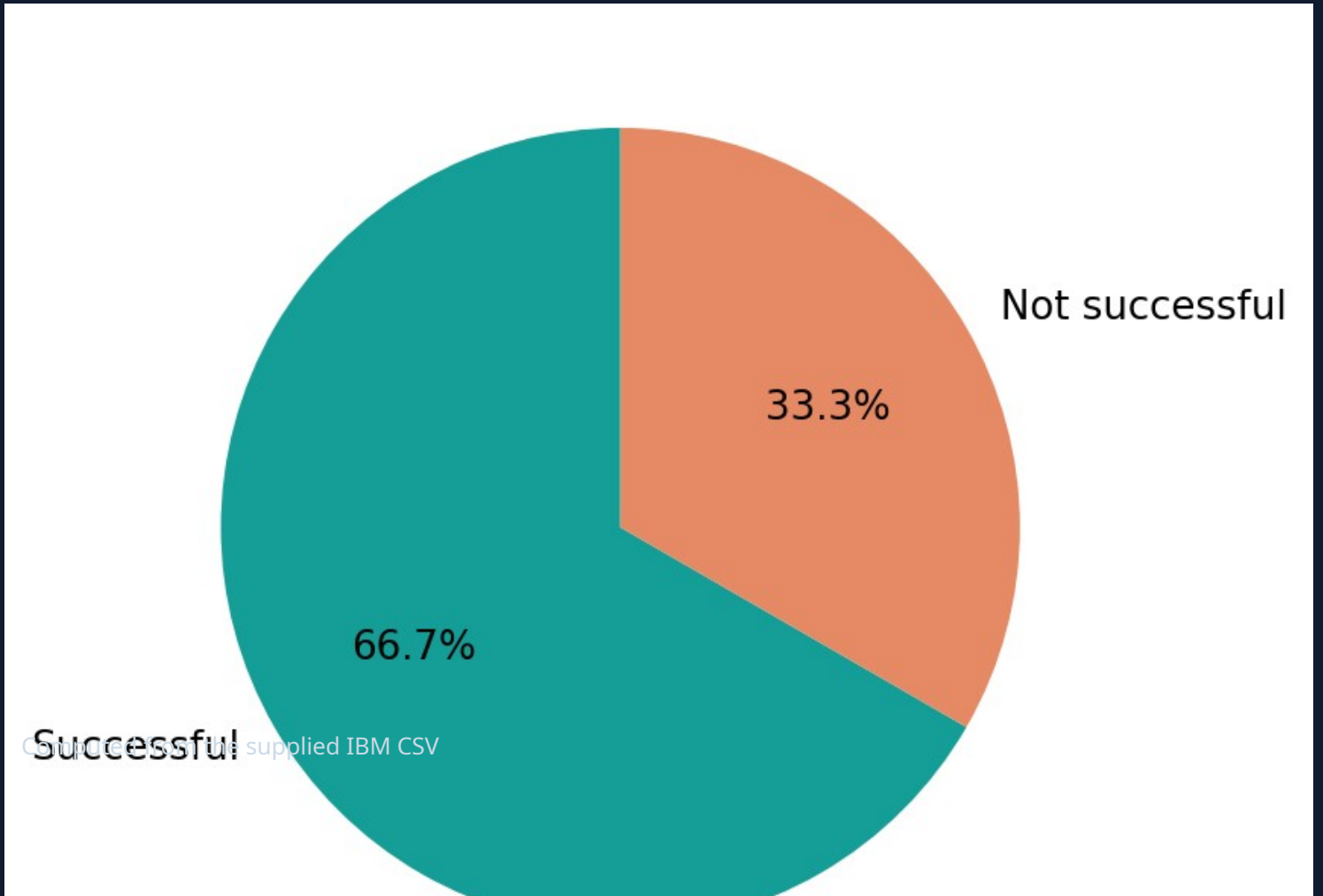
Landing outcome overview

60 successful launches

30 other outcomes

The supplied CSV can power a Dash dropdown, site-filtered pie chart and payload-vs-outcome scatter.

The chart is a static reproduction, not a live Dash screenshot.



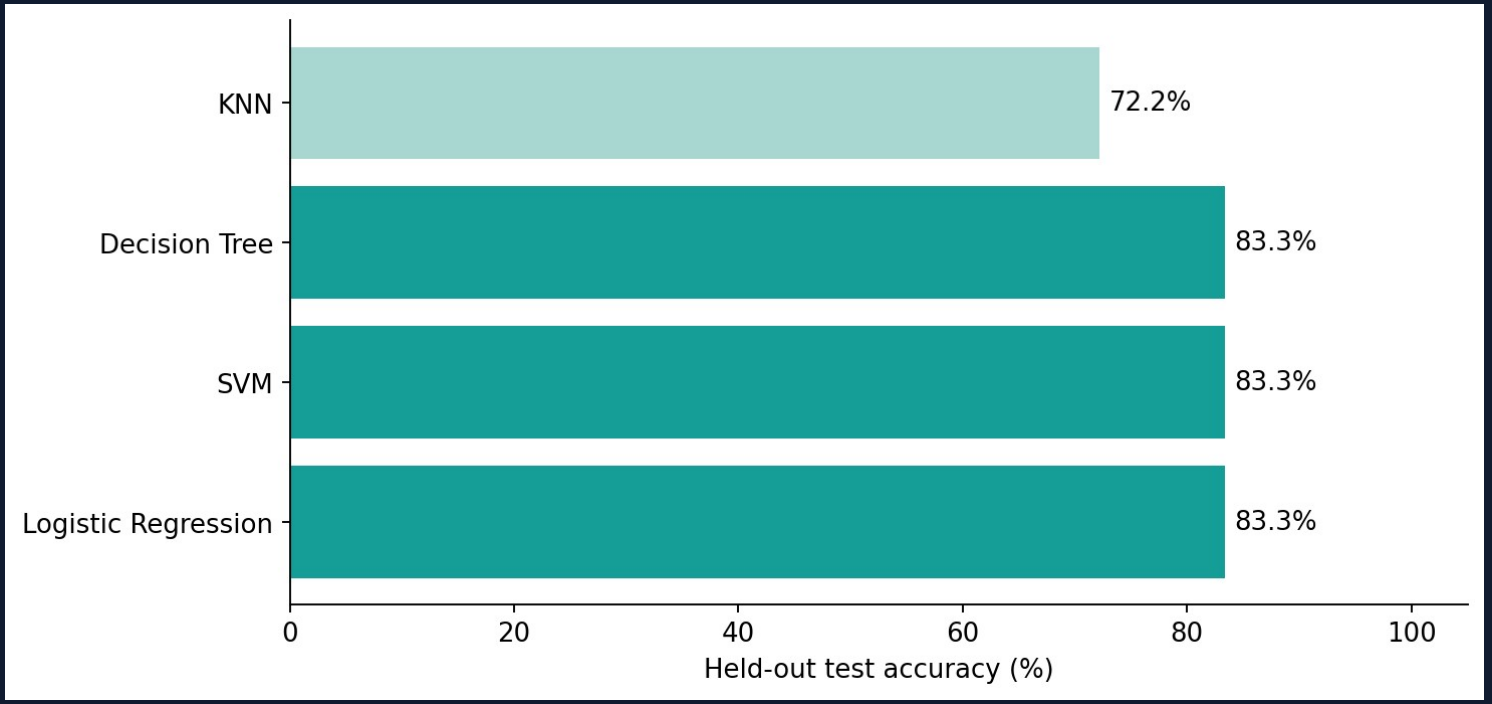
Predictive Modeling: Four Classifiers

SETUP

- 80/20 split: 72 train, 18 test
- random_state = 2
- GridSearchCV: 5 folds
- StandardScaler inside pipeline

MODELS

Logistic Regression, SVM, Decision Tree, KNN



Actual re-run from dataset_part_2.csv and dataset_part_3.csv

Model Evaluation & Confusion Matrix

BEST TEST RESULT: Logistic Regression

Accuracy: 83.33%

Cross-validation score: 82.00%

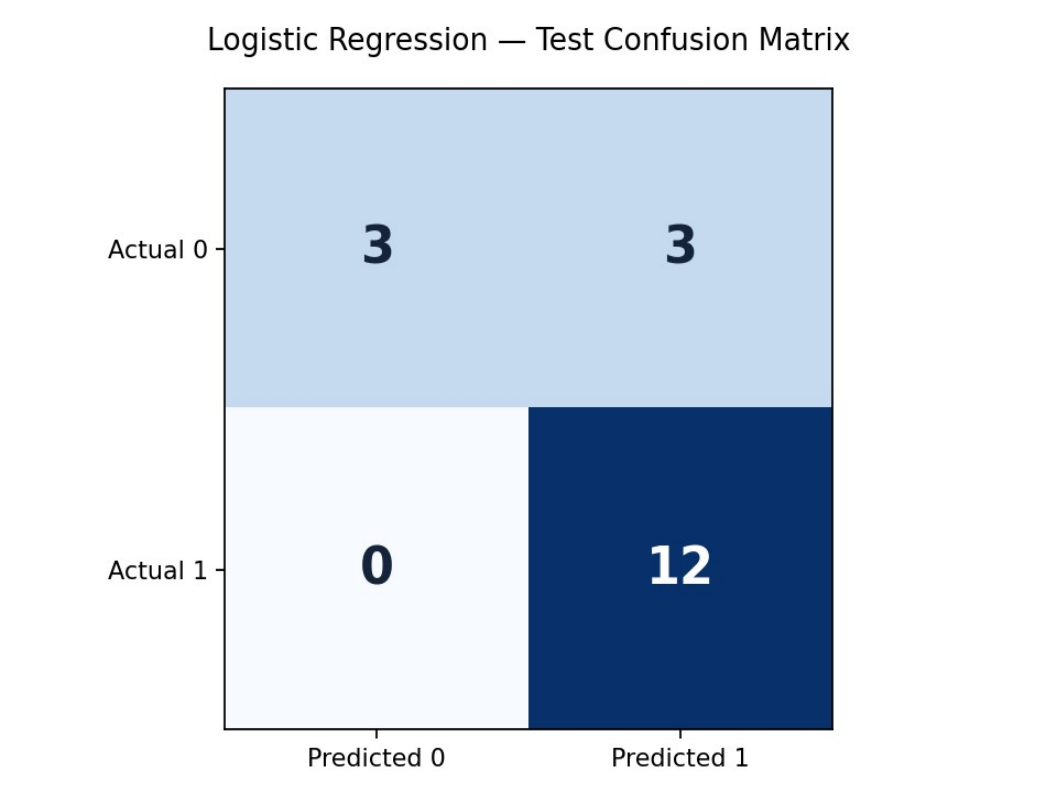
True negatives: 3

False positives: 3

False negatives: 0

True positives: 12

Small test set: 18 cases.



Held-out test predictions

Conclusions & Next Steps

CONCLUSIONS

- The supplied historical sample shows 66.7% successful landings.
- Site, orbit and mission characteristics are associated with different outcomes.
- Reproduced models reach up to 83.3% test accuracy on 18 examples.

LIMITATIONS & NEXT STEPS

- Historical dataset; limited sample size.
- Evaluate newer missions and additional train/test splits.
- Run the original Folium and Dash notebooks to capture interactive screenshots.
- Publish completed notebooks and add real GitHub URLs before submission.